

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

Iventim Solutions GmbH

Hiring Manager

München

Deutschland

Subject: Application for Optical Metrology Engineer

Dear Hiring Manager,

As a physicist and optical engineer with a focus on precision metrology, I am applying for the position of Optical Metrology Engineer in your team. My focus is on leading measurement systems from concept to automated, functional prototypes – a strength I can optimally contribute to your R&D projects.

During my PhD, I designed an automated wavefront measurement system from scratch and controlled it with Python/C++. Daily modeling and tolerance analysis with ZEMAX OpticStudio were as much part of my repertoire as building a Raman spectrometer at IPHT Jena. This end-to-end approach ensures precise measurement results and well-founded design decisions.

Fully integrated into the DACH work culture, I will be available at short notice following the completion of my PhD in May 2026. I look forward to bringing my experience in optical automation to Iventim.

Sincerely,

Juan David Muñoz-Bolaños

Optical Metrology Engineer

Medizinische Universität Innsbruck

Juan David Muñoz-Bolaños

Optical Metrology Engineer

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juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



Professional Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

TECHNICAL COMPETENCIES

Optical Metrology & Simulation

Expert in wavefront sensing, phase retrieval, and ZEMAX OpticStudio tolerancing. Proficient in characterizing aberration sensitivity of high-NA microscope objectives.

System Automation & Control

Experience building C++/Python/LabVIEW control stacks for adaptive optics, scientific cameras (sCMOS), and opto-mechanical stages.

Data Science for Optics

Application of supervised ML and AI algorithms to hyperspectral imaging data. Proficient with NumPy, Pandas, Scikit-learn, and JAX for high-performance computation.

Hardware Integration & Sourcing

End-to-end R&D experience in multi-photon microscope assembly. Competent in technical component specification (Thorlabs, Edmund Optics) and incoming QA.

Process Reliability

Hands-on experience with industrial vision solutions (Halcon) and ensuring measurement traceability via Gage R&R assessments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **Automated Metrology Setups:** Designed and built an automated wavefront sensing system; developed the full software control stack to unify scientific cameras and wavefront modulators.
- **Optical Modeling & Design:** Performed systematic ZEMAX tolerance analysis to optimize the process of custom objective lens characterization and aberrometry.
- **Integration & Alignment:** Achieved diffraction-limited performance in a custom-built nonlinear microscope by precisely aligning galvo scanners, femtosecond lasers, and PMT detectors.
- **Technical Documentation:** Managed technical specification sheets and qualified vendors (Hamamatsu, Thorlabs) for high-precision opto-mechanical integration.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Spectrometer Build:** Handled the end-to-end optical layout and detector calibration of a 1064 nm fiber-probe Raman spectrometer prototype.
- **AI-Driven Data Analysis:** Leveraged custom Python packages for Machine Learning classification on hyperspectral tissue images, accelerating experimental turnaround.
- **Published Research:** Contributed to the development of non-invasive clinical diagnostic tools, results documented in peer-reviewed journals (Applied Sciences, ACS Photonics).

INTECOL S.A.S.
Junior Machine Vision System Engineer

Medellín, Colombia
Mar 2018 – Mar 2019

- **Industrial Machine Vision:** Engineered Halcon inspection systems for beverage production lines; solved critical label and sealing quality issues via custom illumination design.
- **Hardware Commissioning:** Optimized robotic vision for auto-part recognition using smart cameras and tailored illumination methods, directly impacting production efficiency at a car subdivision of Renault.
- **Gage R&R Assessments:** Verified system accuracy and process stability within industrial quality control frameworks.

EDUCATION

Medical University of Innsbruck <i>Ph.D. in Adaptive Optics & Microscopy</i>	Austria Jan 2022 – Present
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Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany 2019 – 2021
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University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia 2012 – 2018
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PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

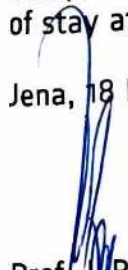
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

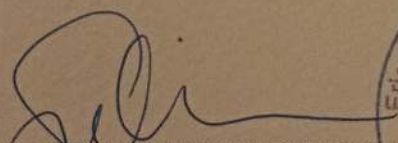
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCÉS, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

OFFICIAL TRANSLATION No. 0439

20 DECEMBER 2018

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passports № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

M LA CRUZ COL

Fecha de expedición / Date of issue

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

Autocidad / Authority

G. ANTIOQUIA

Firma del titolare / Titolare's signature

Dear Family

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